

SALT ANALYSIS

Step-by Step Analysis According to Your Syllabus

Aim of the Experiment: To detect the acidic and basic radical of the supplied inorganic salt by performing dry tests and wet tests and then to identify the salt.

[Explanation:

- A salt is an ionic compound consisting of an anion and cation. The anion of the salt is the acid radical and the cation of the salt is the basic radical.
- Salts are formed by the neutralization reaction of an acid and a base. Acid radical is the ion formed after the removal of H^+ ion from an acid. Basic radical is formed after the removal of OH^- ion from a base.
- Acid radicals included in the 2nd semester syllabus are: CO_3^{2-} , Cl^- , NO_3^- , SO_4^{2-} , S^{2-} .
- Basic radicals included in the 2nd semester syllabus are: Pb^{2+} , Cu^{2+} , Fe^{2+}/Fe^{3+} , Al^{3+} , Zn^{2+} , Ca^{2+} , Ba^{2+} , Mg^{2+} , NH_4^+ , K^+ .
- There are two types of tests to detect the acid radical and basic radical of a salt. These are: Dry Tests and Wet Tests.
- The dry tests are performed in dry conditions, i.e., dry salt and dry test tube is used to perform the test.
- The wet tests are performed by taking the salt solution. The salt solution is prepared in a solvent where the salt is completely soluble.
- Generally, dry tests are indicative, i.e., these tests only indicate that a particular radical may be present in the salt (Exception: Chromyl Chloride Test).
- Whereas, wet tests are confirmatory, i.e., these tests confirm the presence of a radical.]

Physical Characteristics of the Salt:

State: Crystalline / Amorphous

Color: Colored salt / Colorless (white)

Odour: Pungent Smell / Rotten-egg like odour / Odourless

Solubility: Water / Hot Water / dil.HCl / con.HCl / dil.HNO₃ / con.HNO₃ / Aqua-regia

[Explanation: (dil.: dilute ; con.: concentrate)

- **State:** The state or physical appearance of the salt may be crystalline or amorphous. The substances which have definite geometrical shape are called crystalline substances, eg, crystals of sugar. Likewise, the substances which do not have definite geometrical shape are called amorphous substances, eg, powder.

If the salt is crystalline, then it will be soluble in water. Also, there is strong possibility of the presence of these radicals: Cl^- , NO_3^- , NH_4^+ , K^+ .

If the salt is amorphous and powdered in nature; water insoluble, but soluble in dil.HCl, then CO_3^{2-} may be present in the salt

- **Color:** Certain salts have specific colors. By observing the color, information about the salt can be acquired. Some colored salts that are included into your syllabus are: CuSO_4 (dark blue, crystalline), CuCl_2 (dark green, crystalline), CuCO_3 (light green, powdered), FeSO_4 (green, crystalline), $\text{Fe}_2(\text{SO}_4)_3$ (yellow, crystalline).

If the salt is colorless (white), there will be no information regarding the salt.

- **Odour:** The odour of the salt is checked as: Take a small amount of salt (1 part) on your palm, add 2 parts of Na_2CO_3 on it, rub the mixture and take the smell.

For NH_4^+ radical: Pungent smell.

For S^{2-} radical: Rotten-egg like odour.

For other radicals: Odourless.

- **Solubility:** The solubility of the salt is always checked in a definite order. The order is:

Water / Hot Water / dil.HCl / con.HCl / dil. HNO_3 / con. HNO_3 / Aqua-regia

At first, check the solubility in water. If it is soluble in water, then no need to check the solubility in the other solvents. But, if the salt is insoluble in water, then go for hot water. Again, if it is insoluble in hot water, then go for dil.HCl, and so on. The salts included into your syllabus will be surely soluble in either water or dil.HCl.

Aqua-regia is a mixture of two concentrated acids, HCl and HNO_3 in 3:1 ratio.]

Dry Tests for Acid Radical:

Take a small amount of salt (about 0.5g) in a dry test tube and perform the following tests.

Experiment	Observation	Inference
1. Salt + dil. H_2SO_4 + Δ	a) A colorless gas with brisk effervescence. Passing the gas through lime-water, turns it milky. Excess passing of the gas turns lime-water clear again.	CO_3^{2-} may be present.
	b) A colorless gas with rotten-egg like odour. The gas turns lead acetate paper black.	S^{2-} may be present.
	c) No characteristic gas.	CO_3^{2-} , S^{2-} absent.
2. Salt + con. H_2SO_4 + Δ	a) A colorless gas with brisk effervescence. Passing the gas through lime-water, turns it milky. Excess passing of the gas turns lime-water clear again.	CO_3^{2-} may be present.
	b) A colorless gas with rotten-egg like odour. The gas turns lead-acetate paper black.	S^{2-} may be present.
	c) A colorless gas with pungent smell. Hold a glass rod, moistened with NH_4OH , over the mouth of the test tube, dense fumes are produced.	Cl^- may be present.
	d) Reddish-brown fumes after heating.	NO_3^- may be present.
	e) No characteristic gas.	CO_3^{2-} , S^{2-} , Cl^- , NO_3^- absent

Experiment	Observation	Inference
3. Salt + Cu-turnings + con.H ₂ SO ₄ + Δ	a) Dense reddish-brown fumes are produced and the rest of the solution in the test tube turns to blue.	NO ₃ ⁻ may be present.
	b) No reddish-brown gas.	NO ₃ ⁻ absent.
4. <u>Chromyl Chloride Test</u> Salt + K ₂ Cr ₂ O ₇ + con.H ₂ SO ₄ + Δ	a) Red colored gas is produced. Passing the gas through NaOH solution, turns it to yellow.	Cl⁻ confirmed.
	b) No red colored gas.	Cl ⁻ absent.

[N.B. : For each experiment, all the possible observations are given. But for a particular salt, you will get one observation for each experiment. You have to write that observation only.]

Lead-acetate paper: An ordinary filter paper dipped in lead-acetate solution.]

Wet Tests for Acid Radical:

[Prepare the salt solution in that solvent where the salt is completely soluble. Choose the solvent from the solubility of the salt.]

Take a little of salt solution in a test tube and perform the following tests.

Experiment	Observation	Inference
1. Salt solution + AgNO ₃ solution	a) A curdy white ppt. The ppt. is insoluble in dil.HNO ₃ solution, but soluble in NH ₄ OH solution.	Cl ⁻ confirmed.
	b) White ppt. The ppt. is soluble in dil.HNO ₃ solution.	CO ₃ ²⁻ confirmed.
	c) White ppt. The ppt. is soluble on boiling.	SO ₄ ²⁻ confirmed.
	d) No ppt.	Cl ⁻ , CO ₃ ²⁻ , SO ₄ ²⁻ absent.
2. Salt solution + BaCl ₂ solution	a) Heavy white ppt. The ppt. is neither soluble in dil./con.HCl nor dil./con.HNO ₃ .	SO ₄ ²⁻ confirmed.
	b) White ppt. The ppt. is soluble in dil.HCl or dil.HNO ₃ .	CO ₃ ²⁻ confirmed.
	c) No ppt.	CO ₃ ²⁻ , SO ₄ ²⁻ absent.
3. Salt solution + NaOH Solution + Sodiumnitroprusside solution	a) Purple coloration of the solution.	S ²⁻ confirmed.
	b) No purple coloration of the solution.	S ²⁻ absent.
4. <u>Ring Test:</u> Salt solution + Freshly prepared FeSO ₄ solution. Allowed the mixture to cool under water and then add 3-4 drops of con. H ₂ SO ₄ by the side of the test tube.	a) A brown ring is formed between the junction of the two liquids.	NO ₃ ⁻ confirmed.
	b) No brown ring.	NO ₃ ⁻ absent.

[N.B. : For each experiment, all the possible observations are given. But for a particular salt, you will get one observation for each experiment. You have to write that observation only.]

- If the salt is insoluble in water, but soluble in dil.HCl, and get positive observation for CO_3^{2-} radical in the dry test, then wet tests are not performed to confirm the CO_3^{2-} radical. Dry test is considered as the confirmatory test for that radical.]

Dry Tests for Basic Radical:

Experiment	Observation	Inference
1. Heating in a Dry Test Tube		
2. Flame Test		
3. Borax Bead Test		
4. Charcoal Reduction Test		
5. Cobalt Nitrate Test		
6. Ammonium Radical Test		

[N.B. Dry tests for basic radicals give indications about the presence of the basic radical in a salt. You go through your chemistry practical book to know the principle and procedure of each experiment. Also, check the observation and inference for each experiment.]

Wet Tests for Basic Radical:

[N.B. The basic radicals are divided into six (6) groups. Each group contains certain specific basic radicals. These groups are:

Group-I: Ag^+ , Pb^{2+} , Hg_2^{2+}

Group-II: Pb^{2+} , Hg^{2+} , Cu^{2+} , Cd^{2+} , Bi^{3+} , Sb^{3+} , As^{3+}

Group-IIIA: Al^{3+} , $\text{Fe}^{2+}/\text{Fe}^{3+}$, Cr^{3+} , Mn^{2+}

Group-IIIB: Co^{2+} , Ni^{2+} , Zn^{2+} , Mn^{2+}

Group-IV: Ca^{2+} , Ba^{2+} , Sr^{2+}

Group-V: Mg^{2+} , Na^+ , K^+ , NH_4^+

There is no specific reagent to give ppt. for all the radicals of Group V. Therefore, the radicals Na^+ , K^+ , NH_4^+ are placed in a separate group called as: Group Zero.

In some books, you will find that the six groups are named as: Group-1, Group-2, Group-3, Group-4, Group-5, Group-6. You can consider the groups as your convenience. You can Choose Group-I to Group-V; or Group-1 to Group-6.

By performing the wet tests for basic radicals, you can detect which group is present in the salt. After detecting the group, you have to analyze the group ppt. to know which basic radical is present in the supplied salt, as each group contains a number of basic radicals.]

Take a small amount of salt solution in a test tube to perform the following tests:

Experiment	Observation	Inference
1. Salt solution + dil.HCl	a) '.....' ppt.	Group-I present
	b) No ppt.	Group-I absent
2. Above solution + Δ + H ₂ S gas	a) '.....' ppt.	Group-II present
	b) No ppt.	Group-II absent
3. Boiled off H ₂ S gas from the above solution + 1 drop con.HNO ₃ solution (add only if Fe is present in the salt) + NH ₄ Cl (s) + NH ₄ OH solution till alkaline.	a) '.....' ppt.	Group-IIIA present
	b) No ppt.	Group-IIIA absent
4. Above solution + H ₂ S gas	a) '.....' ppt.	Group-IIIB present
	b) No ppt.	Group-IIIB absent
5. Boiled off H ₂ S gas from the above solution + NH ₄ OH + (NH ₄) ₂ CO ₃ (s)	a) White ppt.	Group-IV present
	b) No ppt.	Group-IV absent
6. Above solution + Na ₂ HPO ₄ solution	a) White ppt.	Group-V (Mg ²⁺) present
	b) No ppt.	Group-V (Mg ²⁺) absent

[N.B. For each experiment, all the possible observations are given. But for a particular salt, you will get one observation for each experiment. You have to write that observation only.

- In the '.....' Place, mention the color of the ppt.
- In the test for Group-V, Na₂HPO₄ solution can give ppt. for Mg²⁺ radical only. For other radicals present in Group-V, you have to test separately (zero group= Na⁺, K⁺ and NH₄⁺).
- By performing the wet tests for basic radical, you can detect which group is present in the salt. Each group contains a number of basic radicals. To detect which basic radical is present in the supplied salt, you have to analyze the ppt.]

Tests for K⁺ and NH₄⁺:

Experiment	Observation	Inference
1. Salt solution + Sodiumcobaltinitrite solution	a) Yellow ppt.	K ⁺ or NH ₄ ⁺ present.
	b) No yellow ppt.	K ⁺ , NH ₄ ⁺ absent.
2. Salt solution + Nessler's Reagent	a) Reddish-brown ppt.	NH ₄ ⁺ confirmed.
	b) No reddish-brown ppt.	NH ₄ ⁺ absent.

[N.B. By performing these two tests, you can detect whether the salt contains K^+ or NH_4^+ or not.]

Group-I ppt. Analysis:

The Group-I ppt. is filtered. To the ppt., water is added and then boiled. The ppt. gets dissolved.

Residue	Filtrate
No residue. Ag^+ , Hg_2^{2+} absent.	Divide into two parts. a) 1 st part + K_2CrO_4 (s) Yellow ppt. $\rightarrow$ Pb^{2+} confirmed. b) 2 nd part + KI (s) Yellow ppt. $\rightarrow$ Pb^{2+} confirmed.

Group-II ppt. Analysis:

The group-II ppt. is filtered. Then the ppt. is taken in a test tube and half a test tube of 1:3 HNO_3 is added. Heat gently and then filtered.

Residue	Filtrate	
No residue. Hg^{2+} absent.	Make alkaline by adding NH_4OH . Filter, if ppt. appears.	
	Residue	Filtrate
	(Two Conditions) a) No residue. Pb^{2+} absent. b) White residue + $NaOH$ solution + Δ + Acetic acid + K_2CrO_4 (s) $\rightarrow$ Yellow ppt. $\rightarrow$ Pb^{2+} confirmed.	Blue in color. Filtrate + Acetic acid + $K_4[Fe(CN)_6]$ solution $\rightarrow$ Chocolate brown ppt. $\rightarrow$ Cu^{2+} confirmed.

Group-IIIA ppt. Analysis:

The Group-IIIA ppt. is filtered. The ppt. is taken in a test tube. Add $NaOH$ solution and about 1ml H_2O_2 in it. Boil gently until effervescence stops. Then, filtered.

Residue	Filtrate
a) Reddish-brown residue.	Filtrate + NH_4Cl + Δ $\rightarrow$ White ppt. $\rightarrow$ Al^{3+} confirmed.

The residue is dissolved in dil.HCl and add $K_4[Fe(CN)_6]$ solution. → Deep blue ppt. → Fe^{3+} present. b) No residue. → Fe^{3+} absent.	
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Distinction between Fe^{2+} and Fe^{3+} :

Experiment	Observation	Inference
Salt solution + $K_4[Fe(CN)_6]$ solution	a) Light blue ppt.	Fe^{2+} confirmed.
	b) Deep blue ppt.	Fe^{3+} confirmed

Group-III B ppt. Analysis:

The Group-III B ppt. is filtered. Treat the ppt. with dil.HCl, then warm and filter.

Residue	Filtrate
No residue. Co^{2+} and Ni^{2+} absent.	Filtrate + dil. Acetic acid → Divide into two parts. i) 1st part + H_2S gas → Dirty white ppt. → Zn^{2+} confirmed ii) 2nd part + $K_4[Fe(CN)_6]$ solution → Green ppt. → Zn^{2+} confirmed

Group-IV ppt. Analysis:

The Group-IV ppt. is taken in a test tube and dil. acetic acid is added in it. The ppt. gets dissolved. Divide the solution into two parts.

Experiment	Observation	Inference
i) 1 st part + K_2CrO_4 (s)	a) Yellow ppt.	Ba^{2+} confirmed.
	b) No yellow ppt.	Ba^{2+} absent.
[The 2 nd part is treated in two different ways considering whether Ba^{2+} is present or not.]		
<u>* If Ba^{2+} is absent</u>		
ii) 2 nd part + $(NH_4)_2C_2O_4$ (s)	White ppt.	Ca^{2+} confirmed.
<u>* If Ba^{2+} is present</u>		
ii) 2 nd part is treated with excess K_2CrO_4 and then filtered.	No ppt.	Ca^{2+} absent.
That filtrate + $(NH_4)_2C_2O_4$ (s)		

Group-V ppt. Analysis:

The Group-V ppt, is filtered. The ppt. is taken in a test tube and dil.HCl is added in it. The ppt. gets dissolved.

That solution + NaOH solution → White ppt. → Mg^{2+} confirmed.

[N.B. The above Group ppt. analysis for basic radicals is according to your 2nd semester syllabus.

- Some important questions arise from the analysis of basic radicals. Some of them are given in a tabular form below.

Groups	Group Reagent	Precipitate Formed as:
Group-I	Dil.HCl	Chlorides
Group-II	H_2S gas in presence of dil.HCl	Sulphides
Group-IIIA	NH_4Cl and NH_4OH	Hydroxides
Group-IIIB	H_2S gas in presence of NH_4OH	Sulphides
Group-IV	$(\text{NH}_4)_2\text{CO}_3$	Carbonates
Group-V	Na_2HPO_4	$\text{Mg}(\text{NH}_4)\text{PO}_4$

- By performing the above-mentioned tests serially and systematically, you can easily detect the acidic and basic radical of the supplied salt as well as can identify the salt.

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